

# Elasticity of Intertemporal Substitution

## A Meta-Analysis

Tomas Havranek

Czech National Bank  
Charles University in Prague

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# Why Should We Care About the Elasticity?

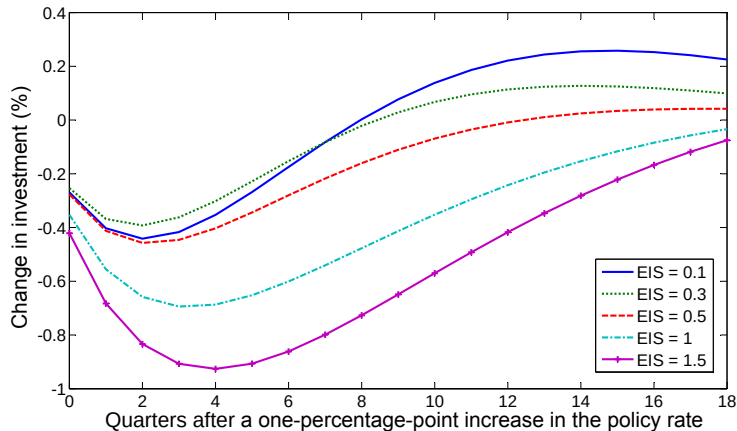
The EIS reflects households' willingness to substitute consumption between time periods in response to changes in the expected real interest rate.

$$u(c) = \frac{c^{1-\frac{1}{EIS}} - 1}{1 - \frac{1}{EIS}}; \quad \text{if } EIS = 1 \Rightarrow u(c) = \log c.$$

Crucial in models involving intertemporal choice:

- monetary policy,
- fiscal policy,
- portfolio choice,
- computing the social cost of carbon emissions, and more.

# The Elasticity Matters.



# Calibrations Vary.

| Study                      | EIS | Outlet                           |
|----------------------------|-----|----------------------------------|
| House & Shapiro (2006)     | 0.2 | American Economic Review         |
| Piazzesi et al. (2007)     | 0.2 | Journal of Financial Economics   |
| Chari et al. (2003)        | 0.2 | Review of Economic Studies       |
| Trabandt & Uhlig (2011)    | 0.5 | Journal of Monetary Economics    |
| Jin (2012)                 | 0.5 | American Economic Review         |
| Rudebusch & Swanson (2012) | 0.5 | American Economic Journal: Macro |
| Smets & Wouters (2007)     | 2/3 | American Economic Review         |
| Bansal & Yaron (2004)      | 1.5 | Journal of Finance               |
| Ai (2010)                  | 2   | Journal of Finance               |
| Barro (2009)               | 2   | American Economic Review         |
| Colacito & Croce (2011)    | 2   | Journal of Political Economy     |

# Different Justifications

| Study                   | EIS | Explanation                                                                                                                                                                             |
|-------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| House & Shapiro (2006)  | 0.2 | p. 1837: "Our calibration is roughly the average estimate in Hall (1988), Campbell and Mankiw (1989), and Barsky et al. (1997)."                                                        |
| Trabandt & Uhlig (2011) | 0.5 | p. 311: "For the intertemporal elasticity of substitution, a general consensus is followed for it to be close to 0.5."                                                                  |
| Smets & Wouters (2007)  | 2/3 | p. 593: "These are all quite standard calibrations."                                                                                                                                    |
| Barro (2009)            | 2   | p. 252: "Because of the shortcomings of the macroeconomic estimates, it is worthwhile to consider microeconomic evidence. The Gruber (2006) analysis is particularly attractive (...)." |

# Call for a Meta-Analysis

Empirical studies produce very different estimates of the EIS.

Browning & Lusardi (1996, p. 1833):

- “It is frustrating in the extreme that we have very little idea of what gives rise to the different findings.”
- “We still await a study which traces all of the sources of differences in conclusions to sample period; sample selection; functional form; variable definition; demographic controls; econometric technique; stochastic specification; instrument definition; etc.”

# Estimating the Elasticity

Researchers usually follow Hall (1988) and use the log-linearized consumption Euler equation:

$$\Delta c_{t+1} = \alpha_i + EIS \cdot r_{i,t+1} + \epsilon_{i,t+1}.$$

- $\Delta c_{t+1}$  is consumption growth at time  $t + 1$
- $r_{i,t+1}$  is the real return on asset  $i$  at time  $t + 1$

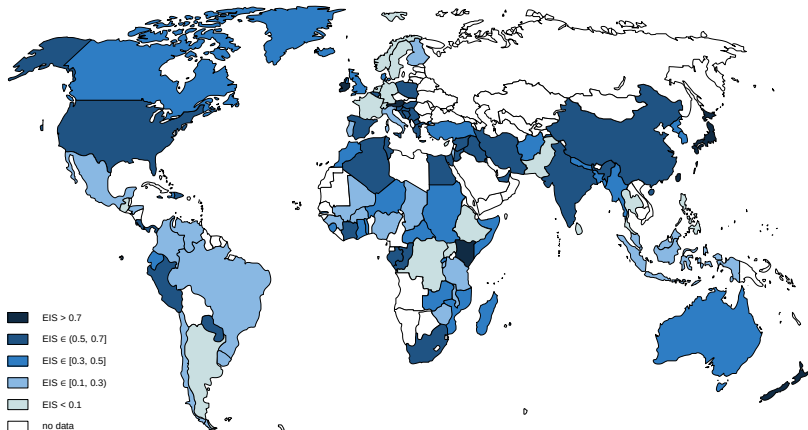
Instruments for  $r_{i,t+1}$  typically include the values of asset returns and consumption growth known at time  $t$ .

# Data Collection

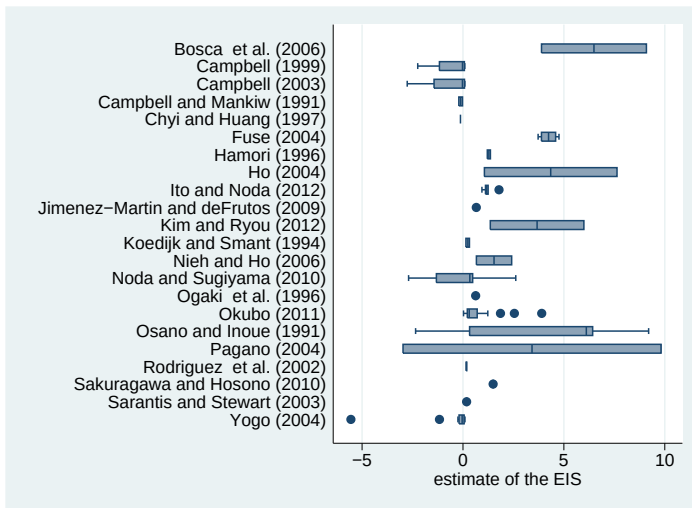
- I find 169 published studies that report estimates of the EIS.
- These studies provide in total 2735 estimates.
- I collect all estimates, their standard errors, and 30 aspects of methodology.
- **The average is 0.5** (0.8 for micro estimates; 0.9 for studies published in top journals).



# The Elasticity Varies Across Countries.



# The Elasticity Varies Across Methods.



# Variables Coded (1)

## Utility

|                        |                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------|
| Epstein-Zin            | =1 if the estimation differentiates between the EIS and the coefficient of relative risk aversion. |
| Habits                 | =1 if habits in consumption are assumed.                                                           |
| Nonsep.<br>durables    | =1 if the model allows for nonseparability between durables and nondurables.                       |
| Nonsep. public         | =1 if the model allows for nonseparability between private and public consumption.                 |
| Nonsep. trad-<br>ables | =1 if the model allows for nonseparability between tradables and nontradables.                     |

## Data

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| No. of house-<br>holds | The logarithm of the number of cross-sectional units used in the estimation (households, cohorts, countries). |
| No. of years           | The logarithm of the number of years of the data period used in the estimation.                               |
| Average year           | The logarithm of the average year of the data period.                                                         |
| Micro data             | =1 if the coefficient comes from a micro-level estimation.                                                    |
| Annual data            | =1 if the data frequency is annual.                                                                           |
| Monthly data           | =1 if the data frequency is monthly.                                                                          |

## Variables Coded (2)

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### *Design*

|                      |                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------|
| Quasipanel           | =1 if quasipanel (synthetic cohort) data are used.                                        |
| Inverse estimation   | =1 if the rate of return is the dependent variable in the estimation.                     |
| Asset holders        | =1 if the estimate is related to the rich or asset holders.                               |
| First lag instrument | =1 if the first lags of variables are included among instruments.                         |
| No year dummies      | =1 if year dummies are omitted in micro studies using the Panel Study of Income Dynamics. |
| Income               | =1 if income is included in the specification.                                            |
| Taste shifters       | The logarithm of the number of controls for taste shifters.                               |

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### *Variable definition*

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| Total consumption | =1 if total consumption is used in the estimation.             |
| Food              | =1 if food is used as a proxy for nondurables.                 |
| Stock return      | =1 if the rate of return is measured as stock return.          |
| Capital return    | =1 if the rate of return is measured as the return on capital. |

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## Variables Coded (3)

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### *Estimation*

|             |                                                           |
|-------------|-----------------------------------------------------------|
| Exact Euler | =1 if the exact Euler equation is estimated.              |
| ML          | =1 if maximum likelihood methods are used for estimation. |
| TSLS        | =1 if two-stage least squares are used for estimation.    |
| OLS         | =1 if ordinary least squares are used for estimation.     |

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### *Publication*

|                  |                                                                                   |
|------------------|-----------------------------------------------------------------------------------|
| SE               | The reported standard error of the estimate of the EIS.                           |
| Publication year | The logarithm of the year of publication of the study.                            |
| Citations        | The logarithm of the number of per-year citations of the study in Google Scholar. |
| Top journal      | =1 if the study was published in one of the top five journals in economics.       |
| Impact           | The recursive RePEc impact factor of the outlet.                                  |

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Funnel asymmetry test:

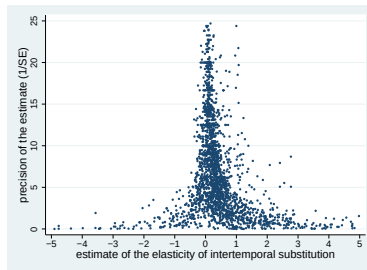
$$\text{EIS}_i = \underbrace{\beta}_{\text{true effect}} + \underbrace{\beta_0 \text{SE}_i}_{\text{publication bias}}$$

|              | FE                  | BE                     | Median                 | IV                 | Micro                | Top                | Country             |
|--------------|---------------------|------------------------|------------------------|--------------------|----------------------|--------------------|---------------------|
| SE           | 2.115***<br>(0.205) | 3.020***<br>(0.573)    | 2.719***<br>(0.397)    | 1.659*<br>(0.850)  | 1.496**<br>(0.717)   | 1.466*<br>(0.825)  | 2.117***<br>(0.216) |
| Constant     | 0.0145<br>(0.00881) | 0.0303***<br>(0.00656) | 0.0322***<br>(0.00893) | 0.0340<br>(0.0363) | 0.174***<br>(0.0554) | 0.171*<br>(0.0887) | 0.0144<br>(0.00928) |
| Observations | 2,735               | 2,735                  | 2,735                  | 2,735              | 512                  | 566                | 2,735               |
| Studies      | 169                 | 169                    | 169                    | 169                | 42                   | 33                 | 169                 |

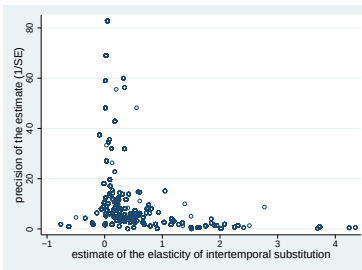
# Negative Estimates Are Underreported.

Negative estimates are published less often than positive estimates with the same precision.

(a) All estimates



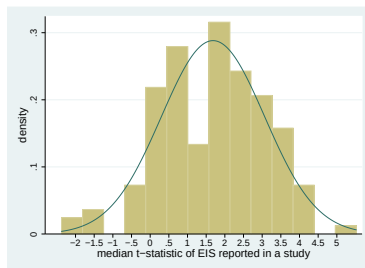
(b) Median estimates



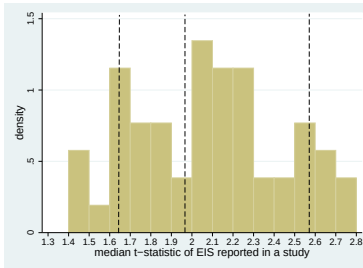
# Marginally Insignificant Estimates Are Underreported.

The probability of reporting increases when the estimate gets another significance star.

(c) Median t-statistics



(d) Median t-statistics around  $t = 2$





# Some Alleged Mistakes in Measurement

- first lags as instruments (time aggregation)
- food as proxy for nondurables (nonseparability)
- omission of time dummies in micro studies (identification would come from time-series variation correlated with consumption)
- assumption of separability between durables and nondurables (nonseparability)
- use of aggregated data (omitted demographic variables)
- log-linearization (endogeneity of higher-order terms)
- use of the rate of return as the dependent variable (weak instruments)
- including non-asset holders (Euler eq. not valid)

# Micro Estimates Are Larger than Macro Estimates.

|               | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  |
|---------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| SE            | 2.465***<br>(0.394)  | 1.926***<br>(0.251)  | 1.864***<br>(0.243)  | 2.109***<br>(0.268)  | 1.975***<br>(0.261)  | 1.961***<br>(0.262)  | 1.809***<br>(0.248)  |
| Micro data    | 0.200***<br>(0.0250) | 0.209***<br>(0.0308) | 0.269***<br>(0.0495) | 0.350***<br>(0.0986) | 0.476***<br>(0.0854) | 0.502***<br>(0.0865) | 0.430***<br>(0.106)  |
| Asset holders | 0.136***<br>(0.0303) | 0.174***<br>(0.0365) | 0.195***<br>(0.0626) | 0.189***<br>(0.0565) | 0.228***<br>(0.0482) | 0.236***<br>(0.0460) | 0.316***<br>(0.0586) |
| Constant      | 0.0237**<br>(0.0109) | 0.00512<br>(0.00322) | -27.52<br>(21.42)    | -37.61*<br>(21.97)   | -26.33<br>(18.44)    | -32.58<br>(22.45)    | -43.89<br>(43.05)    |
| Utility       |                      | Included             | Included             | Included             | Included             | Included             | Included             |
| Data          |                      |                      | Included             | Included             | Included             | Included             | Included             |
| Design        |                      |                      |                      | Included             | Included             | Included             | Included             |
| Variable def. |                      |                      |                      |                      | Included             | Included             | Included             |
| Estimation    |                      |                      |                      |                      |                      | Included             | Included             |
| Publication   |                      |                      |                      |                      |                      |                      | Included             |

# So How Should One Calibrate the EIS?

## Best-practice estimate

I compute a mean estimate conditional on a huge data set, best possible publication characteristics, and lack of major mistakes in measurement.

- Plugging the values to the estimated meta-regression yields  $EIS = 1/3$ .
- The upper confidence bound is  $0.8 \Rightarrow$  calibration of the EIS larger than  $0.8$  is not consistent with the bulk of the empirical evidence.

# Summary

## Main Findings

- 1 People often discard negative or insignificant estimates of the EIS, which drastically biases the mean published estimate upwards.
- 2 Rule of thumb: use  $EIS = 1/3$  for calibrations.
- 3 But: EIS varies a lot across countries.

## Project Website

[www.meta-analysis.cz](http://www.meta-analysis.cz)

## For Further Reading



Stanley, T. D. & C. Doucouliagos (2012): *Meta-Regression Analysis in Economics and Business*.  
Routledge, 1st. edition.



Havranek, T. (2015): *Measuring Intertemporal Substitution: The Importance of Method Choices and Selective Reporting*.  
*Journal of the European Economic Association*, in press.



Havranek, T., R. Horvath, Z. Irsova, & M. Rusnak (2015): *Cross-Country Heterogeneity in Intertemporal Substitution*.  
*Journal of International Economics*: **96(1)**: pp. 100–118.

Reading list on RePEc: Google “meta-analysis in economics.”